

Healthcare Claims Fraud & Payment Risk Analysis

Business Analysis Report

1. Business Problem

Healthcare insurance companies process a large number of claims every day. Some claims may have unusual patterns or potential fraud, which can lead to unnecessary payments and financial losses. This project analyzes claims data to understand payment patterns, identify providers and claim types with higher fraud exposure, and prioritize high-value claims for further investigation.

2. Project Objective

The analysis follows a simple approach:

- Analyse overall claims and payment performance.
- Identify fraud patterns across providers, specialties, insurance and visit types.
- Prioritize high-value fraudulent claims and providers for further investigation.

3. Data & Tools Used

Area	Work Done
Dataset	10,000 healthcare claims with 20 original columns.
Python	Data checking, missing-value handling, date conversion and simple feature creation.
SQL	Business questions on payment, fraud, providers, claim segments, high-value claims and trends.
Power BI	Two-page dashboard for overview, provider risk and claim-level investigation.

The Python notebook created five useful analytical fields: Submission Year, Submission Month, Submission Month Name, Unapproved Amount and Approval Rate.

4. Data Preparation

The data was checked before analysis. There were no cases where Approved Amount was higher than Claim Amount, no negative Claim Amount values, and the final data had no missing values after treatment.

Missing Insurance Type and Provider Specialty values were kept as 'Unknown'. Missing Prior Visits in the last 12 months were filled using the median value. The claim submission date was converted to date format.

5. SQL Analysis

SQL was used to answer practical business questions instead of only practicing SQL syntax. The main questions covered overall payment performance, rejected claims, fraud level, provider risk, specialty risk, insurance and visit-type patterns, high-value fraudulent claims, and monthly fraud trends.

6. Key Findings

Overall payment performance: The dataset contains 10,000 claims. Total submitted claim amount is about \$5.73M, while approved amount is about \$4.76M. The average approval rate is about 84.9%.

Potential fraud level: 829 claims are flagged as potential fraud, which is about 8.3% of all claims. These claims have a total claim amount of about \$821K and approved amount of about \$453K.

Rejected claims: 1,748 claims have a Rejected status. However, the data shows about \$832.8K of approved amount against these rejected claims. This should be reviewed as a possible payment-status data issue or a business-rule case.

Provider risk: Among providers with at least 10 claims, P0086 has the highest fraud rate at about 42.1%, with 16 fraudulent claims out of 38. Other high-rate providers should also be reviewed using both fraud rate and claim volume.

Specialty pattern: General Practice has the highest fraud rate among the listed specialties at about 9.7%, followed by Internal Medicine at about 8.7%.

High-value claims: The highest-value flagged claim is C0000507 from provider P0101, with a claim amount of \$6,590.70 and an unapproved amount of about \$2,319.81.

Important: These results identify claims and providers for review. They do not prove that a claim or provider is fraudulent. Further investigation would be required.

7. SQL Evidence Screenshots

total_claims	total_claim_amount	total_approved_amount	total_unapproved_amount	avg_approval_rate
10000	5728044.06	4755141.57	972902.49	84.88

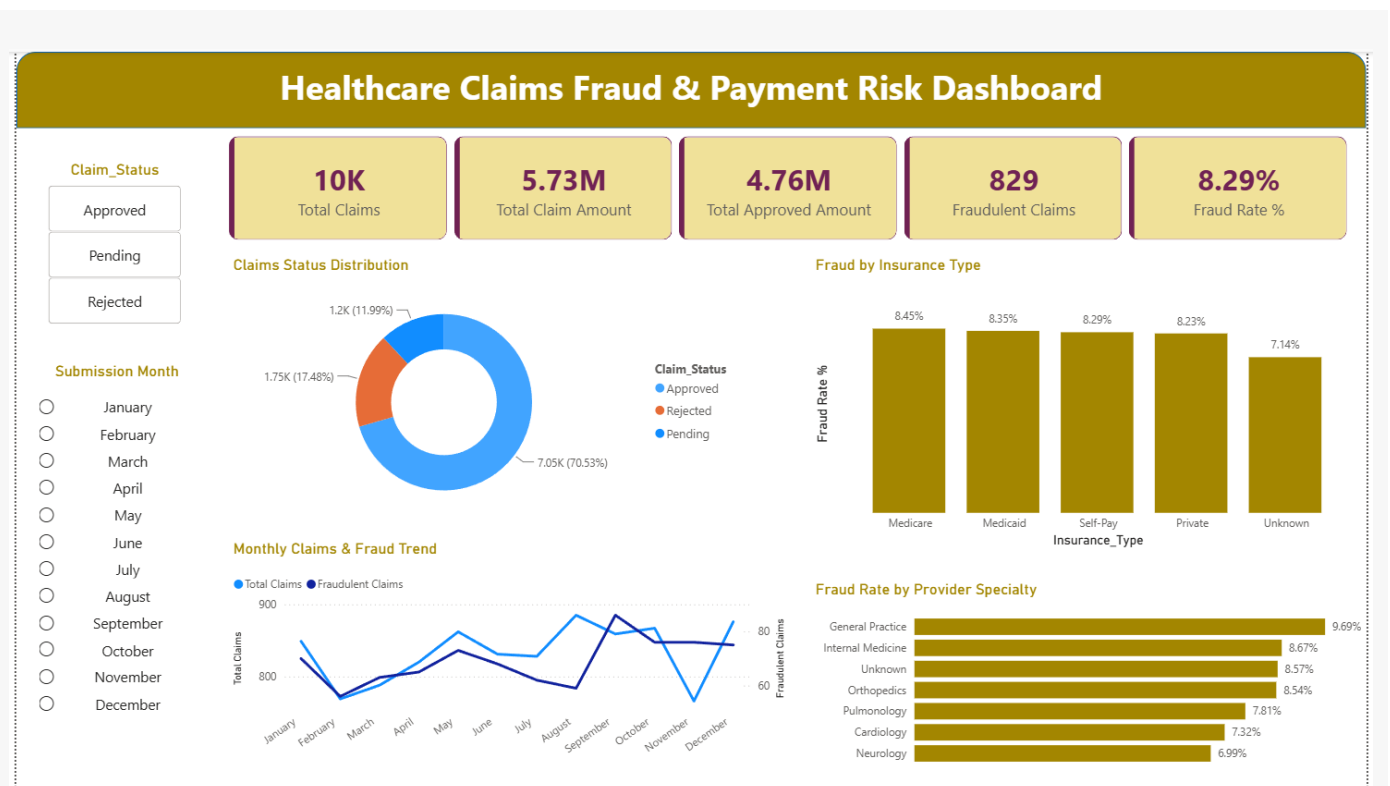
Is_Fraud	claim_count	claim_amount	approved_amount
0	9171	4906561.6	4302613.7
1	829	821482.46	452527.87

Claim_ID	Provider_ID	Claim_Amount	Approved_Amount	Unapproved_Amount	Approval_Rate	Is_Fraud
C0000507	P0101	6590.7	4270.89	2319.81	64.80176612499432	1
C0002682	P0286	3901.94	3100.79	801.15	79.46790570844246	1
C0000589	P0039	3657.89	1195.88	2462.01	32.69316463863047	1
C0008616	P0151	3624.82	1755.51	1869.31	48.4302668822176	1
C0005384	P0289	3469.03	2379.32	1089.71	68.58747257879003	1
C0008764	P0223	3437.05	2134.4	1302.65	62.09976578752128	1
C0004994	P0214	3210.5	1197.24	2013.26	37.29138763432488	1
C0007595	P0091	3188.11	1900.35	1287.76	59.60741630621277	1

8. Power BI Dashboard

The Power BI dashboard has two pages. The first page, 'Claims & Fraud Overview', shows key claim and payment KPIs, claim status, monthly claims and fraud trend, and fraud rate by specialty and insurance type.

The second page, 'Provider Risk & Investigation', focuses on provider-level risk, claim volume versus fraud rate, insurance and visit-type combinations, and the top 10 high-value fraudulent claims.



Healthcare Claims Fraud & Payment Risk Dashboard

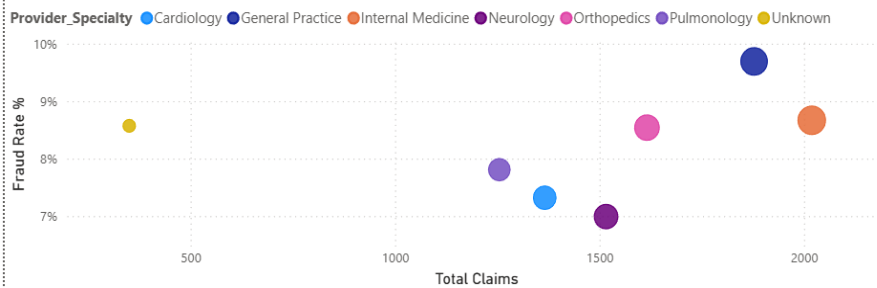
Provider Risk Table

Provider_ID	Total Claims	Fraudulent Claims	Total Claim Amount	Fraud Rate %
P0086	38	16	25,273.33	42.11%
P0159	20	7	10,762.97	35.00%
P0110	45	15	35,672.21	33.33%
P0104	34	11	20,791.88	32.35%
P0223	44	14	35,209.19	31.82%
P0293	33	10	21,560.29	30.30%
P0255	28	8	17,080.72	28.57%
P0027	37	10	23,516.66	27.03%
P0071	32	8	22,532.16	25.00%
P0101	36	9	29,641.86	25.00%
Total	10000	829	57,28,044.06	8.29%

Top 10 High-Value Fraudulent Claims

Claim_ID	Provider_ID	Claim_Amount	Approved_Amount	Approval_Rate	Unapproved_Amount
C0000507	P0101	6,590.70	4,270.89	64.80	2,319.81
C0002682	P0286	3,901.94	3,100.79	79.47	801.15
C0000589	P0039	3,657.89	1,195.88	32.69	2,462.01
C0008616	P0151	3,624.82	1,755.51	48.43	1,869.31
C0005384	P0289	3,469.03	2,379.32	68.59	1,089.71
C0008764	P0223	3,437.05	2,134.40	62.10	1,302.65
C0004994	P0214	3,210.50	1,197.24	37.29	2,013.26
C0007595	P0091	3,188.11	1,900.35	59.61	1,287.76
C0006322	P0238	3,094.60	1,272.51	41.12	1,822.09
C0001895	P0131	2,957.86	2,060.05	69.65	897.81

Provider Risk: Claim Volume vs Fraud Rate



Fraud Rate by Insurance & Visit Type

Insurance_Type	Emergency	Inpatient	Outpatient	Total
Medicare	9.03%	7.85%	8.42%	8.45%
Medicaid	8.40%	8.28%	8.37%	8.35%
Self-Pay	9.21%	8.16%	7.48%	8.29%
Private	7.06%	9.10%	8.53%	8.23%
Unknown	8.26%	4.59%	8.33%	7.14%
Total	8.43%	8.23%	8.21%	8.29%

9. Business Recommendations

- Review providers with high fraud rates and enough claim volume before taking any action.
- Prioritize high-value flagged claims for manual review because they can have higher financial impact.
- Review rejected claims that still show approved payment amounts to understand whether this is a data or process issue.
- Monitor specialties and insurance/visit-type combinations with higher fraud rates over time.
- Use the dashboard as a monitoring and investigation-support tool, not as proof of fraud.

10. Conclusion

This project shows how a Data Analyst can take healthcare claims data, prepare it in Python, answer business questions using SQL, and present the results in Power BI. The main focus is on understanding payment

performance, finding areas with higher potential fraud exposure, and helping the business decide where further review may be useful.

Project flow: Python → SQL → Power BI